

CCS2300 - Data Structures and Algorithms Group Assignment Guidelines

Assignment Type: Group Project with Practical Demonstration and Viva Voce

Weightage: 40% of the Final Grade

- **Project Progress Video (20%)**
- **Final Project Report, Source Code Submission, Practical Demonstration, and Individual Viva Voce (20%)**

Group Size: Exactly 6 members per group (No less and no more than 6 members)

Assignment Title:

Design and Development of a Data Structures-Based Smart Management System

Scenario

Each group will be assigned a real-world problem domain after the completion of group registration. The assigned topic may be from areas such as:

- Library Management System
- Hospital Management System
- Student Management System

Students are required to design, develop, test, and evaluate a software application based on the assigned topic. The developed system must demonstrate the practical application of the data structures and algorithms covered in the module.

The implementation must be carried out using **Java** as the programming language.

Note: Groups are not permitted to select their own project topics. Topics will be allocated after the group registration process is completed. Once assigned, groups must complete the project based on the allocated topic.

Assignment Requirements

The system must incorporate the following Data Structures and Algorithms.

Task 1 - Linear Data Structures (ILO2, ILO4)

Implement and demonstrate:

- Arrays
- Singly or Doubly Linked List
- Stack
- Queue

Example:

- Arrays for storing fixed-size collections such as categories, departments, room information, or predefined configurations.
- Linked List for storing and managing dynamic records.
- Stack for undo/recent activity operations
- Queue for service requests, bookings, appointments, or waiting lists.

Deliverables

- Source code
- Explanation of implementation
- Screenshots and testing evidence
- Complexity analysis (time and space complexity)

Task 2 - Hashing and Sets (ILO2, ILO4)

Implement:

- Hash Table
- Set ADT

The system should:

- Store and retrieve records efficiently
- Handle collisions using an appropriate collision resolution technique.
- Demonstrate duplicate prevention using the Set ADT.

Deliverables

- Source code
- Hash function explanation
- Collision handling method
- Testing evidence
- Performance observations
- Complexity analysis (time and space complexity)

Task 3 - Searching and Sorting Algorithms (ILO1, ILO3)

Implement and compare:

Searching

- Linear Search

Sorting

- Bubble Sort
- Selection Sort
- Insertion Sort
- Merge Sort
- Quick Sort

Students must:

- Implement the assigned searching and sorting algorithms.
- Compare the execution times of the implemented algorithms.
- Discuss the efficiency and suitable use cases of each algorithm.
- Justify the selection of algorithms for the developed system.

Deliverables

- Source code
- Test data
- Comparison table
- Performance evaluation
- Discussion
- Complexity analysis (time and space complexity)

Task 4 - Algorithm Complexity Analysis (ILO3)

Each group member must analyze the computational complexity of the data structures and algorithms implemented under their assigned component.

The analysis should include:

- Best-case time complexity
- Average-case time complexity
- Worst-case time complexity
- Space complexity
- Justification of the findings based on theoretical analysis and experimental observations.

Deliverables

- Complexity analysis report
- Performance observations
- Justification of findings

Task 5 -Trees (ILO2, ILO4)

Implement:

- Binary Search Tree (BST)
- AVL Tree

Demonstrate:

- Insert
- Delete
- Search
- Traversals
 - Inorder
 - Preorder
 - Postorder

Deliverables

- Source code
- Sample outputs
- Screenshots
- Complexity analysis (time and space complexity)

Task 6 - Graphs (ILO2, ILO4)

Implement a graph-based feature relevant to the selected system.

Examples:

- Route finding
- Relationship mapping
- Dependency tracking
- Network visualization

Demonstrate:

- Graph representation
- Traversal algorithms
 - BFS
 - DFS

Deliverables

- Source code
- Graph visualization
- Test results
- Complexity analysis (time and space complexity)

Group Task Distribution

Each group must contain exactly **6 members**.

The workload must be divided equally as follows:

Member	Responsibility
Member 1	Arrays, Linked Lists, Linear Search, and complexity analysis
Member 2	Stack, Queue, Bubble Sort, and complexity analysis
Member 3	Binary Search Tree (BST), Selection Sort, and complexity analysis
Member 4	AVL Tree, Insertion Sort, and complexity analysis
Member 5	Graphs (Graph Representation, BFS, DFS), Merge Sort, and complexity analysis
Member 6	Hash Tables, Set ADT, Quick Sort, and complexity analysis

Note: Although each member has primary responsibility for specific components, **every group member must actively contribute to:**

- System design
- Coding and implementation
- Module integration
- System testing and debugging
- Performance evaluation
- Documentation
- Practical demonstration
- Final presentation

Every student is expected to understand the overall system architecture and the interaction between all implemented data structures and algorithms.

Documentation Requirements

The final report should include:

1. Introduction
2. Problem Description
3. System Design
4. Data Structures Used
5. Algorithms Used
6. Complexity Analysis
7. Testing and Validation
8. Screenshots and Results
9. Challenges Encountered
10. Conclusion
11. References

Submission Requirements

Project Progress Video (20%)

Project Progress Video Submission

Submission Week: Week 10

Duration: 8-10 Minutes per Group

Purpose

The Project Progress Video is intended to evaluate the progress of the project prior to the final submission. The video should demonstrate the work completed up to Week 10, including the implementation of the data structures covered during the lectures, current system functionality, and each member's contribution.

The assessment will be used to monitor project development, verify participation of all group members, and provide formative feedback before the final submission and individual viva voce.

Video Requirements

Each group must submit a single video covering the following sections:

1. Project Introduction

The group must briefly introduce:

- Project title
- Assigned problem domain
- Objectives of the proposed system

2. System Design

The group must present:

- Overall system architecture
- Main system modules and functionalities
- System workflow

3. Data Structures and Algorithms

Students should demonstrate the implementation completed up to Week 10, including:

- Arrays
- Linked Lists
- Stack
- Queue
- Binary Search Tree (BST)
- AVL Tree
- Graphs
- Hash Tables
- Set ADT

Students should explain:

- Why each data structure was selected.
- How each data structure is used within the system.
- Current implementation status.
- Preliminary testing conducted.

4. Progress Demonstration

The group must demonstrate the current implementation of the system, including:

- Features completed so far
- Current system functionality
- Screenshots or live execution of the application
- Preliminary testing evidence

5. Individual Contribution

Every group member must appear in the video and explain:

- **Their assigned component.**
- **Progress achieved.**
- **Data structures implemented.**
- **Complexity considerations of the implemented component.**
- Challenges encountered and solutions applied.

6. Future Development Plan

The group must discuss:

- **Remaining implementation of searching and sorting algorithms.**
- **Complexity analysis of the remaining algorithms.**
- **System integration activities.**
- **Final testing and evaluation.**
- **Documentation completion.**
- Planned milestones before the final submission.

Project Progress Video Evaluation

Criteria	Marks
Project Understanding and Problem Analysis	05
System Design and Architecture	10
Progress Demonstration	40
Application and Justification of Data Structures and Algorithms	20
Individual Contributions and Technical Explanation	20
Professionalism, Organization, and Clarity of Presentation	05
Total	100

Important Notes

- Every group member must appear and actively participate in the video presentation.
- The video must clearly demonstrate the actual progress achieved by Week 10.
- **Submission of the Project Progress Video is mandatory.**
- Groups that fail to provide sufficient evidence of progress may receive reduced marks.
- Failure to submit the Project Progress Video will result in a mark of zero for this assessment component.

Since the Project Progress Video is conducted in **Week 10**, students are only expected to demonstrate the implementation of the data structures and algorithms covered up to Week 10. The searching and sorting algorithms introduced in **Weeks 11 and 12** will be implemented, analyzed, and evaluated as part of the final project submission and demonstrated during the final practical assessment and individual viva voce.

Final Project Submission (20%)

Submission Requirements

Submit the following through the LMS:

1. Source Code
2. Project Report (PDF)
3. Group Contribution Sheet

Assessment Criteria

Criteria	Marks
Linear Data Structures Implementation (Linked List) with all the deliverables	08
Stacks and Queues with all the deliverables	08
Hashing and Sets with all the deliverables	08
Searching, Sorting and Complexity Analysis with all the deliverables	08
Trees Implementation with all the deliverables	08
Graph Implementation with all the deliverables	08
Documentation Quality (with all the required submissions)	02
Total	50

Practical Demonstration and Viva Voce

A **mandatory practical demonstration and individual viva voce** will be conducted after the submission.

The viva is part of the assignment evaluation and will be used to verify each student's contribution, technical knowledge, and overall understanding of the project.

Viva Requirements

Each student must:

- Demonstrate the component primarily assigned to them.
- Explain the implementation details of their assigned component.
- Explain the data structures and algorithms used throughout the project.
- Demonstrate the execution of the complete system.
- Answer questions related to any component of the project, including those implemented by other group members.
- Explain how the different data structures and algorithms are integrated within the system.
- Discuss the design decisions, complexity considerations, and testing procedures used in the project.

Note: Although each member is assigned a specific responsibility, every student is expected to have a comprehensive understanding of the entire project. During the viva, questions may be asked from any section of the project, not only the component assigned to the student.

Important: Attendance at the viva is mandatory. Any student who fails to participate in the viva examination will be awarded **0 marks** for the entire assignment, regardless of their contribution to the project.

Individual Viva Evaluation

Criteria	Marks
Understanding of Assigned Component	15
Understanding of Other Project Components	15
Code Explanation, Logic, and Practical Demonstration	15
Knowledge of Data Structures and Algorithms	05
Total	50

Important: A student's viva mark will be awarded based on their individual performance. Students who are unable to explain their assigned work or demonstrate an adequate understanding of the overall project may receive reduced marks regardless of the group's overall performance.

Academic Integrity

- All submitted work must be original.
- Plagiarism, AI-generated submissions without proper understanding, copied code, or inability to explain the implementation during the viva will result in mark deductions or a zero mark according to institutional regulations.
- Students are expected to fully understand and justify every component of the submitted work.